STACKS, SHEAVES ON ALGEBRAIC STACKS: PROPOSITION-QC-DERIVED-COMPLETE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5136-5149 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 5151-5174, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 26 of queue_algebraic-stacks-sheaves.json (pb=1107, stmt 5136-5149, proof 5151-5174).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 04TM: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0AIF: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0AIA: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07E8: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZE: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZK: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZG: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5174–5174. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let S be a scheme. Let $X = Spf(A)$ where A is an an adic Noetherian topological S-algebra with ideal of definition I, see More on Algebra, Definition 07E8 and Formal Spaces, Definition 0AIF. Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ the be category fibred in sets associated to the functor X, see Categories, Example 04TM. Then $QC(\mathcal{X})$ is canonically equivalent to the category $D_{comp}(A, I)$ of objects of $D(A)$ which are derived complete with respect to I.*

Proof. Recall that $X = \operatorname{colim} \operatorname{Spec}(A/I^n)$ as an fppf sheaf. An object of $\mathcal{X}_{affine}$ is the same thing as an affine scheme $U = \operatorname{Spec}(R)$ with a given morphism $f : U \to X$. By Formal Spaces, Lemma 0AIA there exists an $n \geq 1$ such that f factors through the monomorphism $\operatorname{Spec}(A/I^n) \to X$. Consider the full subcategory $\mathcal{C} \subset \mathcal{X}_{affine}$ consisting of the objects $\operatorname{Spec}(A/I^n) \to X$. By the remarks just made and Differential Graded Sheaves, Lemma 0GZE restriction to $\mathcal{C}$ is an exact equivalence $QC(\mathcal{X}) \to QC(\mathcal{C}, \mathcal{O}|_{\mathcal{C}})$. For simplicity, let us assume that $I^n \neq I^{n+1}$ for all $n \geq 1$. Then $(\mathcal{C}, \mathcal{O}|_{\mathcal{C}})$ is isomorphic as a ringed site to the ringed site $(\mathbf{N}, (A/I^n))$, see Differential Graded Sheaves, Section 0GZG. Hence we conclude by Differential Graded Sheaves, Proposition 0GZK. □